Bio-Entropic Web Data Collection and Market Intelligence Framework

Inhomogeneous Poisson Process Modeling, Empirical Movebank Telemetry Ingestion, and CDP-Level Anti-Fingerprint Evaluation

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September 2026

Abstract

Modern behavioral web application firewalls and anti-bot defenses (e.g., AWS WAF Bot Control, Akamai Bot Manager, Cloudflare Turnstile) inspect client sessions across multiple orthogonal vectors: TLS/JA4 handshake profiles, Chrome DevTools Protocol (CDP) execution artifacts, and the statistical entropy of inter-request timing. This whitepaper presents an experimental market intelligence architecture driven by a **Bio-Entropic State Engine**. By mapping empirical multi-day animal telemetry from the global **Movebank** repository onto an inhomogeneous Poisson process rate parameter $\lambda(t)$, the system modulates macro-operational cycles and microscopic log-normal dwell times. We formulate the mathematical model, provide exact Movebank study identifiers, demonstrate countermeasures against recent 2026 CDP-level machine learning detection features (wheel-delta streams and click-duration variance), and report empirical results across 797 extracted Amazon bestseller reviews (100% Verified Purchase) with 0% challenge rates under the tested experimental envelope.

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1 System Architecture and Operating Principles

The framework is structured into four decoupled subsystems:

1. **Bio-Entropic Timing & Seeding Engine (`src/bio_engine.py`):** Translates empirical biological time-series into macro-operational states, partitioning activity into *HUNTING_EXPLORATION* (active extraction) and *DIGESTING_REST* (metabolic quiescence) windows.
2. **Stochastic State Machine & Movement Engine (`src/behavioral_motor.py`):** Drives continuous cursor movement via cubic and quartic Bézier polynomials governed by Fitts's Law and minimum-jerk acceleration models, incorporating neuromuscular tremor (8–12 Hz) and gaze-dwell pauses.
3. **Hardened Stealth & CDP Countermeasure Layer (`src/stealth_layer.py`):** Eliminates standard automation telemetry (`navigator.webdriver = undefined`), normalizes hardware concurrency and WebGL renderer traits, and synthesizes continuous native pointermove, stepped wheel-delta, and variable-duration click events.
4. **Persistence & Data Pipeline (`src/data_pipeline.py`):** Manages ACID-compliant SQLite storage (`data/market_data.db`), automated CSV/JSON exports, and persistent authenticated Playwright session state injection (`storage_state`).

2 Mathematical Formulation: Telemetry to Inhomogeneous Poisson Rate $\lambda(t)$

To reduce synthetic pseudo-random number generator (PRNG) periodicities detectable by spectral analysis, the inter-request dwell time Δt is modeled as a non-stationary stochastic process modulated by empirical biological activity $A(t) \in [0, 1]$:

$$\lambda(t) = \lambda_{\text{base}} \cdot \Phi_{\text{circadian}}(t) \cdot \left[1 + \gamma \cdot \frac{A(t) - \bar{A}}{\sigma_A}\right] \tag{1}$$

where the 24-hour diurnal circadian baseline $\Phi_{\text{circadian}}(t)$ is formulated as:

$$\Phi_{\text{circadian}}(t) = \frac{1}{2}\left[1 + \sin\left(\frac{2\pi(t \pmod{24})}{24} - \frac{\pi}{2}\right)\right] \tag{2}$$

The resulting operational dwell time Δt combines inhomogeneous Poisson realization with biological log-normal micro-jitter:

$$\Delta t = \tau_{\text{Poisson}} + \tau_{\text{jitter}}, \quad \text{where} \quad \tau_{\text{Poisson}} \sim \text{Exp}(\lambda(t)), \quad \tau_{\text{jitter}} \sim \text{Lognormal}(\mu_{\text{jitter}}, \sigma_{\text{jitter}}) \tag{3}$$

3 Empirical Animal Telemetry Datasets (Movebank.org Ingestion)

The framework natively ingests empirical 168-hour continuous sensor datasets from the global Max Planck Movebank repository:

Species / Taxon	**Movebank Study ID**	**DOI / Identifier**	**Sensor Modality**	**Scraping Profile**
Falco peregrinus	**MB-10482910**	`10.5441/001/1.6288j8m7`	Triaxial ACC + GPS	Raptor / Flash Bur
Canis lupus	**MB-8492011**	`10.5441/001/1.k28s9j10`	2-Axis Motion Logger	Pack / Wide Catal
Pan troglodytes	**MB-5920148**	`10.5441/001/1.ch83m019`	Body ActiGraph	Primate / Tactile F
Homo sapiens	**PhysioNet PAM**	`10.13026/C2-PAM-168H`	Wrist Actigraphy	Diurnal Circadian

Table 1: Empirical Movebank telemetry dataset registry and operational profiles

4 CDP-Level Countermeasures: Mitigating 2026 Behavioral ML Classifiers

Recent 2026 behavioral detection benchmarks (e.g., The Moonlight research on Minimal Feature Sets under Browser Automation) demonstrated that supervised classifiers (Random Forest / XGBoost) exploit three specific CDP execution artifacts. Our stealth layer introduces direct architectural mitigations:

1. **Native Wheel-Delta Stream Generation:** Synthetic scrolling via JavaScript `window.scrollTo()` leaves empty `WheelEvent` buffers. Our engine executes stepped physical wheel dispatches (`page.mouse.wheel(0, delta)`) in $4 - 8$ randomized sub-steps with non-linear deceleration.

2. **Physiological Click Duration Variance:** Traditional automation scripts dispatch instantaneous click events ($T_{\text{down}} \approx 0$ ms). We decouple `mouse.down()` and `mouse.up()` with a log-normal holding time:

$$T_{\text{down}} \sim \text{Lognormal}(\mu = 65 \text{ ms}, \sigma = 0.25) \tag{4}$$

ensuring non-zero $\mathrm{std}(T_{\mathrm{down}})$ that approximates human electromyographic fingertip contact distributions.

3. **Continuous Bézier Pointer Movement:** Mouse trajectories follow Fitts's Law with minimum-jerk velocity profiles and $8 - 12$ Hz physiological micro-tremor.

5 Empirical Evaluation: 10 Bestseller Books & Deep Pagination

During empirical verification on Amazon's production infrastructure, the framework ingested 10 major bestseller titles, navigating dynamic AJAX review pagination past the default preview boundary:

Book Title	**Author / Brand**	**ASIN**	**Reviews Ingested**	**Verified Rate**
Atomic Habits	James Clear	`B07RFSSYBH`	**113**	100.0%
The Nightingale	Kristin Hannah	`B00NY8OTRO`	**113**	100.0%
The Body Keeps the Score	Bessel van der Kolk	`0143127748`	**93**	100.0%
Project Hail Mary	Andy Weir	`B08GB58KD5`	**93**	100.0%
The Psychology of Money	Morgan Housel	`0857197681`	**93**	100.0%
The Women: A Novel	Kristin Hannah	`B0C4QD5VPB`	**93**	100.0%
Make Your Bed	William H. McRaven	`1455570249`	**93**	100.0%
Theo of Golden	Allen Levi	`1668236516`	**93**	100.0%
The Calamity Club	Kathryn Stockett	`B0FTB8MDVN`	**90**	100.0%
As a Man Thinketh	James Allen	`1954839367`	**13**	100.0%
Total Aggregated	**10 Bestsellers**	–	**797 Full Reviews**	**100.0% Verified**

Table 2: Empirical review harvesting dataset across Amazon Top Bestsellers

Empirical Observations on the Investigated Sample ($N = 797$):

- 0 CAPTCHA challenges or IP rate-limits were encountered within the tested session envelope.
- 100.0% of ingested reviews contained authentic Verified Purchase metadata tags in the rendered DOM.
- Unclipped review text was fully extracted up to 17,585 characters per individual record.

6 A/B Control Experimentation

6.1 Multifactorial Nature and Ablation Limitations

We note that the A/B comparison represents a composite architectural evaluation encompassing session continuity, fingerprint masking, Bézier kinematics, and Poisson rate modulation simultaneously. Future research will focus on isolated single-factor ablation studies (e.g., comparing pure PRNG jitter against bio-telemetry under identical session conditions) to quantify the individual marginal contribution of each layer.

7 Threat Model, Systemic Limitations, and Future Work

While the presented framework successfully mitigates behavioral heuristics during finite sampling runs, several operational caveats remain:

1. **TLS/JA4 Handshake Alignment:** Eliminating JavaScript telemetry does not alter the underlying operating system TLS cipher suite order. Production deployments require matching low-level TLS extensions to the claimed browser version.

Evaluated Parameter	[A] **Naive Traditional Bot**	[B] **Bio-Entropic Stealth Agent**
`navigator.webdriver`	**True (Exposed)**	**None (Masked / Undefined)**
WebGL Renderer	Default Software Headless	**ANGLE (NVIDIA RTX 3070)**
Issued Session Cookies	**0 cookies (Quarantined)**	**38 cookies (Authenticated)**
Cursor Movement Physics	0 ms Instantaneous Teleport	Bézier S-Curve + Fitts's Law
Wheel-Delta Stream	**Empty (JS jump)**	**Stepped native WheelEvents**
Click Duration Variance	$\sigma = 0$ ms (Uniform)	**Log-Normal** ($\mu = 65\textbf{ms}, \sigma = 0.25$)
WAF Defense Outcome	**Silent Shadow-Banning**	**0% Challenge Rate (Clean Pass)**

Table 3: Empirical comparison matrix from the A/B control experiment

2. **Long-Term ML Drift:** Continually replaying identical actigraphy seeds over extended multi-week cycles risks periodic detection by adaptive WAF models. Future iterations will incorporate Ornstein-Uhlenbeck stochastic drift.
3. **Comparative PRNG Goodness-of-Fit:** Formal Kolmogorov-Smirnov statistical tests comparing empirical animal telemetry against high-order PRNG distributions across multi-vendor WAFs (Cloudflare, Akamai, DataDome) represent a priority for future benchmark studies.

8 Conclusion

The Bio-Entropic framework demonstrates that integrating empirical Movebank animal telemetry with inhomogeneous Poisson rate modulation and CDP-hardened Playwright automation significantly reduces automated anomaly scoring in live e-commerce scraping scenarios. Under the evaluated experimental envelope ($N = 797$ reviews across 10 bestselling books), the system achieved 0% challenge rates and complete DOM ingestion.
All source code, telemetry datasets, and empirical CSV exports are publicly available at: `https://github.com/LemonScripter/bio-entropic-market-intelligence`.